

Database Lab – Week’s Contents

Lab 1: NUMBER Datatype

Task 1: Create a table products with columns:

- product_id (NUMBER(4))
- product_name (VARCHAR2(20))
- price (NUMBER(7,2))

Solution:

```
CREATE TABLE products ( product_id NUMBER(4), product_name VARCHAR2(20), price
NUMBER(7,2) ); INSERT INTO products VALUES (1001, 'Laptop', 85000.50); INSERT INTO
products VALUES (1002, 'Mouse', 1200.75); INSERT INTO products VALUES (1003,
'Keyboard', 2500.00); SELECT * FROM products;
```

Lab 2: DATE Datatype

Task 2: Create a table employees with emp_id, emp_name, hire_date and insert 3 employees.

Solution:

```
CREATE TABLE employees ( emp_id NUMBER(4), emp_name VARCHAR2(20), hire_date DATE );
INSERT INTO employees VALUES (101, 'Ali', TO_DATE('01-JAN-2022','DD-MON-YYYY'));
INSERT INTO employees VALUES (102, 'Sara', TO_DATE('15-FEB-2023','DD-MON-YYYY'));
INSERT INTO employees VALUES (103, 'Ahmed', TO_DATE('25-SEP-2025','DD-MON-YYYY'));
-- Display employees hired in 2023 SELECT * FROM employees WHERE TO_CHAR(hire_date,
'YYYY') = '2023';
```

Lab 3: Conversion Functions

Task 3: Display system date in format Day, DD-Mon-YYYY

```
SELECT TO_CHAR(SYSDATE, 'Day, DD-Mon-YYYY') AS "Formatted Date" FROM dual;
```

Task 4: Convert string '12345' into number and add 55

```
SELECT TO_NUMBER('12345') + 55 AS result FROM dual;
```

Task 5: Convert string '2025-09-25' into DATE

```
SELECT TO_DATE('2025-09-25', 'YYYY-MM-DD') AS "Converted Date" FROM dual;
```

Lab 4: REGEXP_LIKE

Task 6: Find employees whose names start with A

```
SELECT * FROM employees WHERE REGEXP_LIKE(emp_name, '^A');
```

Task 7: Find employees whose names end with a

```
SELECT * FROM employees WHERE REGEXP_LIKE(emp_name, 'a$');
```

Task 8: Find employees whose name contains only alphabets

```
SELECT * FROM employees WHERE REGEXP_LIKE(emp_name, '^[A-Za-z]+$');
```

Task 9: Find employees hired in 2022 using REGEXP

```
SELECT * FROM employees WHERE REGEXP_LIKE(TO_CHAR(hire_date, 'YYYY'), '2022');
```

■ Commands List

Command	Description
NUMBER(p,s)	Defines numeric data type (precision, scale).
DATE	Stores date & time values.
TO_CHAR()	Converts number/date to string.
TO_NUMBER()	Converts string to number.

TO_DATE()	Converts string to date.
SYSDATE	Returns current system date.
REGEXP_LIKE(col, pattern)	Matches column values with regex pattern.